

CSA1715-ARTIFICIAL INTELLIGENCE

LAB PROGRAM- 20

18-08-2026

20. PROLOG PROGRAM FOR PLANETS DATABASE

AIM

To write and execute a Prolog program to create a database containing information about the planets in the Solar System. The program demonstrates the representation of facts and retrieval of planetary information using queries.

ALGORITHM

1. Start the program.
2. Define facts for the planets in the Solar System.
3. Store information such as planet name, order, and type.
4. Use queries to retrieve the required planet information.
5. Display the result.
6. Stop the program.

PROLOG CODE

% Planets database

planet(mercury, 1, terrestrial).

planet(venus, 2, terrestrial).

planet(earth, 3, terrestrial).

planet(mars, 4, terrestrial).

planet(jupiter, 5, gas_giant).

planet(saturn, 6, gas_giant).

planet(uranus, 7, ice_giant).

planet(neptune, 8, ice_giant).

SAMPLE INPUT

?- planet(earth, Position, Type).

SAMPLE OUTPUT

Position = 3,

Type = terrestrial.

RESULT

Thus, the Prolog program for representing a planets database was successfully implemented. Planetary information was stored as facts and retrieved correctly using Prolog queries.